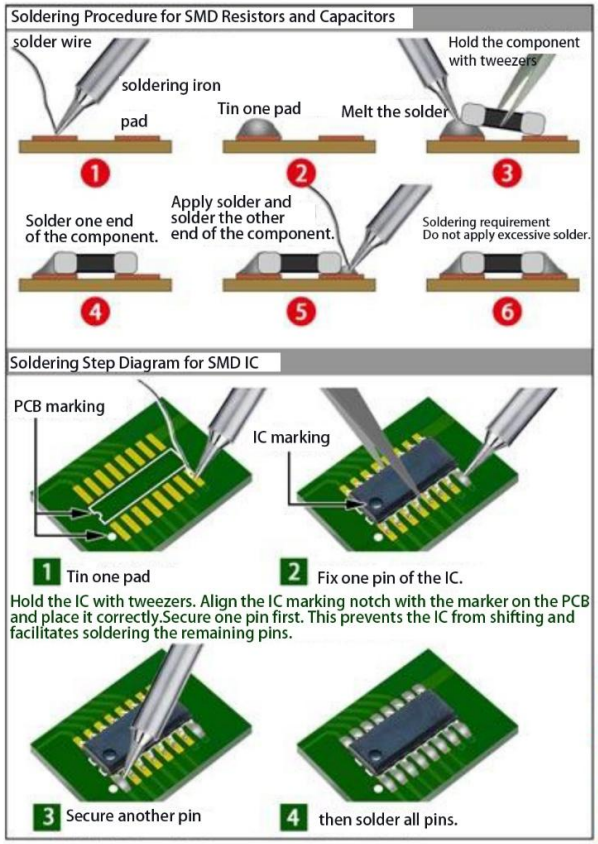
LED Spectrum Installation Tutorial

****SMD Soldering Procedure****



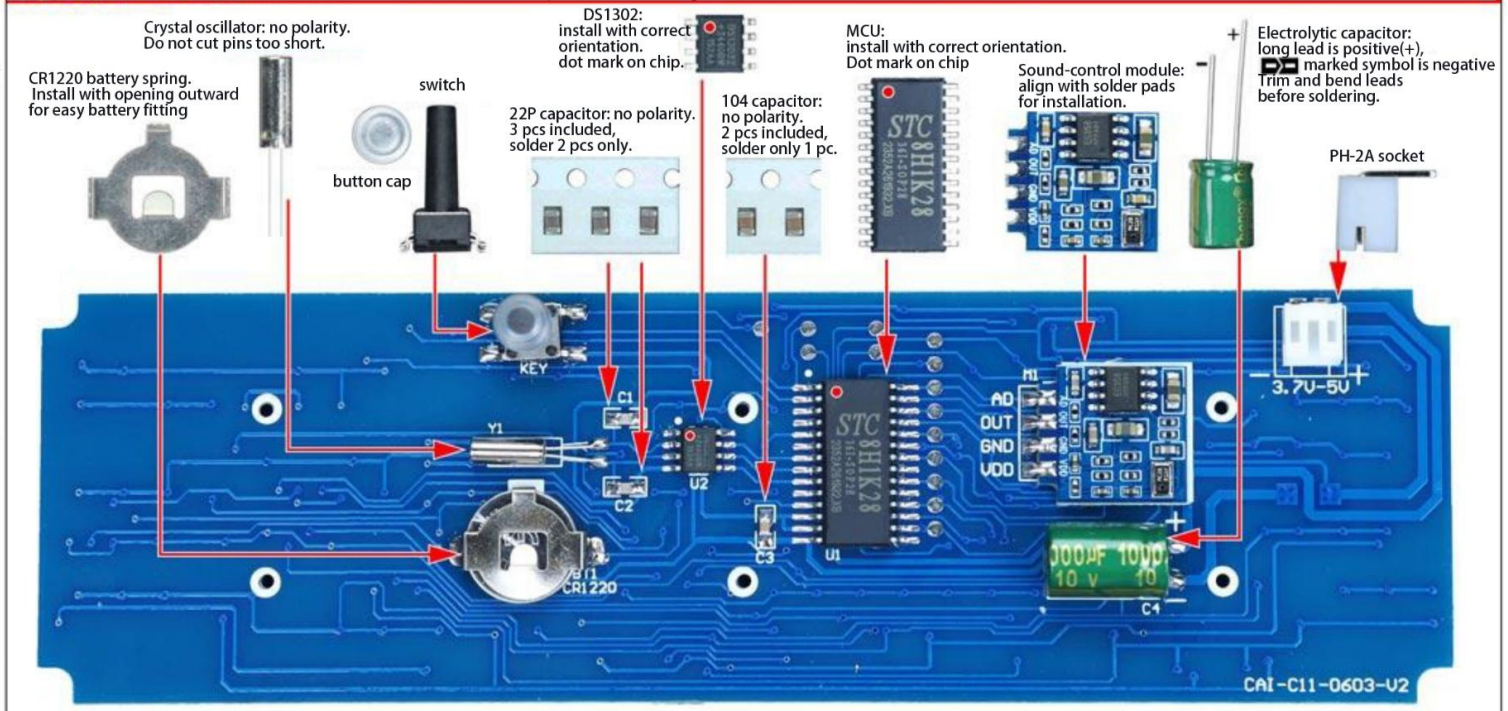
Scan QR code to view soldering video.

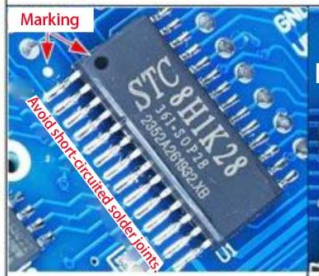
Installation Notes

1. The LED beads are pre-soldered and have passed testing.
2. Handle with care to avoid damaging the LED beads.
3. Do not solder with a hot air gun. High temperature may cause the LEDs on the reverse side to fall off.
4. No solder paste allowed (it may cause timing failure. Rosin is acceptable).

Special Notes:

Hot air gun soldering is forbidden; high heat may detach back-side LED beads.
No solder paste (may cause timing failure. Rosin can be used).





1. Install IC: align chip mark with PCB mark (see figure).
2. Avoid short circuits between IC pins; keep solder joints clean and qualified.



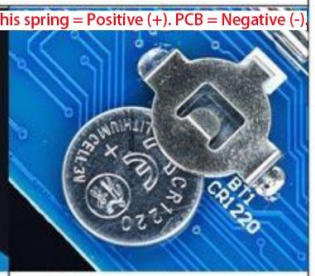
1. Leave crystal pins slightly longer.
2. Mount 22pF capacitors at positions C1 and C2.
3. All three are 22pF, non-polarized. Solder paste is prohibited (rosin flux is acceptable).



1. Confirm polarity, bend pins and trim excess. Leave pins slightly longer.
2. Align to +/- pads, horizontal mount. Add more solder for stronger joints.



1. Tactile switch is non-polarized. Align with pads for soldering and ensure centered placement.
2. Fit the transparent keycap. Avoid impact damage to the switch after assembly.



1. Solder two joints of battery spring securely. Opening outward for easy battery fitting.
2. Fit battery per above diagram



Fix soldered PCB to grid housing using 4 small screws. Tighten properly; do not over-tighten.



Connect power port with terminal wire. Red wire = positive, match (+). Do not overforce during plugging to prevent connector drop-off.Glue if detached.



Snap on the transparent rear housing and secure with screws.Tighten screws properly; do not over-tighten.

Package list



SOP28 IC(MCU) x1

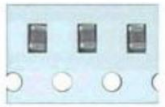

keycap x1

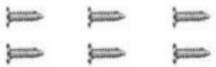

SOP IC(DS1302) x1


PH-2A Bent Pin Socket ×1


0805 SMD Capacitor (104) ×2


1.7x6mm Screw ×4

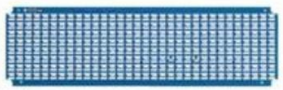

0805 SMD Capacitor (22P) ×3


1.4x4.5mmScrew ×6


32768Hz Crystal Oscillator ×1


PH2.0 Terminal Wire ×1


Sound Control Module x1


Circuit Board x1
(LED SMD LEDs pre-soldered)


CR1220 Battery Spring x1


0832 Dot Matrix Housing x1


1000uF Electrolytic Capacitor x1


Transparent Rear Housing x1


6x6x15 switch x1


USB to Type-C Power Cable x1

Operating Instructions

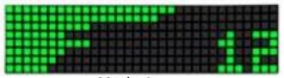
Display Mode Setting

Setting method: Short press the key (tap once).
Each press switches to the next mode.


Mode 1:
Buoy + Vertical Bar Real-time Display Mode


Mode 2:
Level Display + Vertical Bar Real-time Display


Mode 3:
Vertical Bar Real-time Display Mode


Mode 4:
Horizontal Real-time Display Mode
+ Numeric Level Display


Mode 5:
Horizontal Real-time Display Mode


Mode 6:
Buoy + Horizontal Three-group Combined Bar Display


Mode 7:
Vertical Real-time Display Mode


Mode 8:
Vertical Central Axis Real-time Display Mode


Mode 9:
Buoy + Vertical Bar Real-time Display Mode


Mode 10:
Vertical Buoy Real-time Display Mode


Mode 11:
Buoy + Horizontal Two-group Combined Real-time Mode

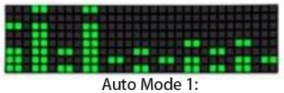

Mode 12:
Horizontal Shift Half-wave Mode


Mode 13:
Horizontal Shift Full-wave Mode


Mode 14:
Half-wave Mode + Numeric Level Display


Mode 15:
Vertical Real-time Mode + Full-wave Mode


Mode 16:
Buoy + Horizontal Single Combined Real-time Display


Auto Mode 1:
Sequential cycle from Mode 1 to Mode 16


Auto Mode 2:
Random switching among Mode 1 to Mode 16

Restore Factory Settings

Disconnect the power first. Hold down the key, then reconnect the power, and then release the key.The LEDs will flash several times regularly, then all light up (this is the LED test).Press the switch once; two scrolling numbers will appear.(These are seconds. If the numbers do not count, check the installation of the crystal oscillator and DS1302 components.)Press the switch again to exit the test. Factory reset is completed.

Other Function Settings

1. Long press the key (hold for about 1 second) to enter the setting interface.
2. Short press the key (tap lightly once) to modify the value.
Long press the key again to enter the next setting interface.
3. To exit settings: wait 8 seconds for automatic exit, or cut off power to exit.
4. Restore Factory Settings: Hold down the key, then power on.
All LEDs will flash several times regularly, then release the key.
The numbers will start counting. Press the key once more to exit.

Time Setting:

Set 24-hour format or 12-hour format.
Set hour value.
Set minute value.



Sensitivity Setting:

1-12: The higher the value, the higher the sensitivity, and the lower the adaptable signal strength.
00: Spectrum display off



Transition Duration Setting:

Time interval for mode switching in auto mode
Setting range: 5 seconds to 20 seconds



Display Mirror

0: Output in default mode
X: X-axis mirrored output relative to default
Y: Y-axis mirrored output relative to default



Display Speed

The higher the value,
the higher the screen refresh rate



Brightness Adjustment

Brightness levels 1 – 8
(Default: maximum brightness, level 8)



Clock Display

1: Enable clock display function
0: Disable clock display function



Delay Setting
The waiting time to switch to clock display when no audio signal is present.
(Number × 3) seconds.
Example: 2 × 3 = 6 seconds.



Night Mode

0: Disable this function; 1: Enable this function.
21 – 07 means dim display from 9 p.m. to 7 a.m.

